Delivering Military Advantage through multi-national geospatial interoperability

DGIWG-NWP-RADAR-001

Edition 1.0

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NWP Radar Profile

CONTENTS

I. ABSTRACT

SECURITY CLASSIFICATION: NATO RESTRICTED (NR) (NATO)

This document defines the NWP Radar Profile, an OGC API — Environmental Data Retrieval (EDR) Part 3 Service Profile. It specifies normative requirements and conformance tests for server implementations conforming to this profile.

II. KEYWORDS

The following are keywords to be used by search engines and document catalogues.

NWP, earth observations, cube, radius, items, datetime, precipitation_amount, radar

III. PREFACE

This document was prepared by the submitting organizations.

IV. SECURITY CONSIDERATIONS

No security considerations have been made for this document.

V. SUBMITTERS

All questions regarding this document should be directed to the editor or the submitters:

Table — Submitters

NAME	AFFILIATION
Austria	Institute of Military Geography
France	Institut National de l'Information Géographique et Forestière (IGN)
Finland	Finnish Defence Forces
Germany	Bundeswehr Geoinformation Centre (BGIC)
NATO	NATO Maritime GEOMETOC Centre of Excellence
United Kingdom	Met Office (MO), UK Strategic Command, DSTL
United States	National Geospatial-Intelligence Agency (NGA)

VI. POINT OF CONTACT

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1. SCOPE

This standard defines the NWP Radar Profile. It specifies requirements and conformance tests for implementations of this profile.

2. CONFORMANCE

Conformance with this standard shall be checked using the Abstract Test Suite in Annex A.

The following conformance classes are defined:

- http://www.dgiwg.org/std/edr/1.0/conf/nwp_radar

3. NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Mark Burgoyne, David Blodgett, Charles Heazel, Chris Little, Open Geospatial Consortium: OGC 19-086r6, *OGC API — Environmental Data Retrieval Standard*. Open Geospatial Consortium (2023). http://www.opengis.net/doc/IS/ogcapi-edr-1/1.1.0.

OGC API — EDR Part 3: Service Profiles (draft)

4. TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this standard.

This document also uses terms defined in OGC API — EDR Part 1: Core.

4.1. service profile

A named, referenceable set of constraints and additional requirements applied to an OGC API — EDR implementation to satisfy a specific community or mission need.

SOURCE: OGC API — EDR Part 3: Service Profiles (draft)

4.2. coverage

A feature that acts as a function returning values from its range for any position within its spatiotemporal domain.

SOURCE: OGC 19-086r6, OGC API — Environmental Data Retrieval Standard

5. REQUIREMENTS

REQUIREMENTS CLASS 1	
IDENTIFIER	`http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
CONFORMANCE CLASS	Conformance class A.1: `http://www.dgiwg.org/std/edr/1.0/conf/nwp_radar`
TARGET-TYPE	NWP Radar Profile Profile Standard
NORMATIVE STATEMENTS	Requirement 1: `/req/nwp_radar/spatial-extents-crs` Requirement 2: `/req/nwp_radar/profile-bbox-coordinates` Requirement 3: `/req/nwp_radar/profile-trs-values` Requirement 4: `/req/nwp_radar/profile-vrs-values` Requirement 5: `/req/nwp_radar/profile-keywords` Requirement 6: `/req/nwp_radar/profile-links` Requirement 7: `/req/nwp_radar/profile-query-types` Requirement 8: `/req/nwp_radar/items-endpoint` Requirement 9: `/req/nwp_radar/cube-endpoint` Requirement 10: `/req/nwp_radar/radius-endpoint` Requirement 11: `/req/nwp_radar/profile-nwp-parameters` Requirement 12: `/req/nwp_radar/http-methods`

REQUIREMENT 1	
IDENTIFIER	`/req/nwp_radar/spatial-extents-crs`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	The coordinates for the bounding box SHALL be defined in the CRS EPSG:4326 (WGS84) with geographical latitude, then longitude, in decimal degrees.
A	The service bbox SHALL have a geometry defined in WGS84.
B	The service bounding box coordinates SHALL be returned with geographical latitude, then longitude.
C	The service bounding box coordinates SHALL be returned in decimal degrees.

REQUIREMENT 2	
IDENTIFIER	`/req/nwp_radar/profile-bbox-coordinates`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	The coordinates for the 2D bounding box SHALL be defined in the CRS84 EPSG:4326 (WGS84) for the extent [-180, -90, 180, 90].
A	The 2D bounding box boundary SHALL be for the extent [-180, -90, 180, 90].
B	The coordinate reference system for the bounding box SHALL be EPSG:4326 (WGS84).

DRAFT

REQUIREMENT 3	
IDENTIFIER	`/req/nwp_radar/profile-trs-values`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	Temporal Reference System (trs) values SHALL be presented in Gregorian calendar Zulu datetime (UTC) in 24 hour clock with a precision to the second according to the time format specified in RFC 3339.
A	The service trs SHALL be in Gregorian calendar Zulu datetime (UTC) in 24 hour clock with a precision to the second.
B	The service trs SHALL be the time format specified in RFC 3339.

REQUIREMENT 4	
IDENTIFIER	`/req/nwp_radar/profile-vrs-values`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	Vertical Reference System (vrs) values SHALL be height in metres, with an unambiguous direction of increase declared.
A	The service vrs SHALL be in height metres.
B	The service vrs SHALL be EPSG:3855.
C	The service vrs SHALL use the vertical datum EGM2008 geoid as the zero-height reference (orthometric height relative to mean sea level).
D	The service SHALL declare extent.vertical.positive as “up” or “down” so vertical interval ordering (e.g. pressure levels ordered top-to-bottom vs. bottom-to-top) is unambiguous.

REQUIREMENT 5	
IDENTIFIER	`/req/nwp_radar/profile-keywords`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	The following keywords SHALL be included in the collection response: “cube”, “radius”, “items”, “precipitation_amount”, “radar”.
A	The collection keywords array SHALL include “cube”, “radius”, and “items”.
B	The collection keywords array SHALL include “precipitation_amount” and “radar”.

REQUIREMENT 6	
IDENTIFIER	`/req/nwp_radar/profile-links`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	All links objects SHALL have href, rel, title and type attributes.
A	Each link object SHALL include a non-empty href attribute.
B	Each link object SHALL include rel, title and type attributes.

REQUIREMENT 7	
IDENTIFIER	`/req/nwp_radar/profile-query-types`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	The following data queries SHALL be supported: “cube”, “items”, “radius”.
A	The collection SHALL advertise cube, items and radius data queries.

REQUIREMENT 8	
IDENTIFIER	`/req/nwp_radar/items-endpoint`

REQUIREMENT 8	
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	The service SHALL provide a /collections/nwp_radar_precipitation/items endpoint.
A	The service SHALL return CoverageJSON FeatureCollection and GRIB2, GeoTIFF, HDF5, NetCDF4, PNG file formats.
B	The service SHALL return a default file format of GeoTIFF.
C	Each feature SHALL include precipitation_amount property.
D	Each feature SHALL include datetime property in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z.
E	The items data queries SHALL return CoverageJSON that complies with the OGC CoverageJSON standard — https://docs.ogc.org/cs/21-069r2/21-069r2.html#_df6ce66a-7369-49cb-b59f-47a8c5d424ca
F	The items data queries SHALL return GeoTIFF which complies with the DGIWG 108 GeoTIFF for GeoRef Imagery Standard.
G	The items data queries SHALL return GRIB2 which complies with the WMO Manual On Codes — https://library.wmo.int/records/item/35625-manual-on-codes-volume-i-2-international-codes.
H	The items data queries SHALL return HDF5 which complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.
I	The items data queries SHALL return NetCDF4 which complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.

REQUIREMENT 9	
IDENTIFIER	`/req/nwp_radar/cube-endpoint`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	The service SHALL provide a /collections/nwp_radar_precipitation/cube endpoint.
A	The service SHALL return CoverageJSON FeatureCollection and GRIB2, GeoTIFF, HDF5, NetCDF4, PNG file formats.
B	The service SHALL return a default file format of GeoTIFF.
C	Each feature SHALL include precipitation_amount property.
D	Each feature SHALL include datetime property in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z.

REQUIREMENT 9	
E	The cube data queries SHALL return CoverageJSON that complies with the OGC CoverageJSON standard — https://docs.ogc.org/cs/21-069r2/21-069r2.html#_df6ce66a-7369-49cb-b59f-47a8c5d424ca
F	The cube data queries SHALL return GeoTIFF which complies with the DGIWG 108 GeoTIFF for GeoRef Imagery Standard.
G	The cube data queries SHALL return GRIB2 which complies with the WMO Manual On Codes — https://library.wmo.int/records/item/35625-manual-on-codes-volume-i-2-international-codes.
H	The cube data queries SHALL return HDF5 which complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.
I	The cube data queries SHALL return NetCDF4 which complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.

REQUIREMENT 10	
IDENTIFIER	`/req/nwp_radar/radius-endpoint`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	The service SHALL provide a /collections/nwp_radar_precipitation/radius endpoint.
A	The service SHALL return CoverageJSON FeatureCollection and GRIB2, GeoTIFF, HDF5, NetCDF4, PNG file formats.
B	The service SHALL return a default file format of GeoTIFF.
C	Each feature SHALL include precipitation_amount property.
D	Each feature SHALL include datetime property in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z.
E	The radius data queries SHALL return CoverageJSON that complies with the OGC CoverageJSON standard — https://docs.ogc.org/cs/21-069r2/21-069r2.html#_df6ce66a-7369-49cb-b59f-47a8c5d424ca
F	The radius data queries SHALL return GeoTIFF which complies with the DGIWG 108 GeoTIFF for GeoRef Imagery Standard.
G	The radius data queries SHALL return GRIB2 which complies with the WMO Manual On Codes — https://library.wmo.int/records/item/35625-manual-on-codes-volume-i-2-international-codes.
H	The radius data queries SHALL return HDF5 which complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.
I	The radius data queries SHALL return NetCDF4 which complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.

REQUIREMENT 11	
IDENTIFIER	`/req/nwp_radar/profile-nwp-parameters`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	The following parameters SHALL be supported: datetime, precipitation_amount.
A	The parameter datetime SHALL be in Gregorian calendar Zulu datetime (UTC) in 24 hour clock with a precision to the second.
B	The parameter datetime SHALL be the time format specified in RFC 3339.
C	The parameter precipitation_amount SHALL return values in kg m-2 units.

REQUIREMENT 12	
IDENTIFIER	`/req/nwp_radar/http-methods`
INCLUDED IN	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
STATEMENT	The service SHALL be configured to return data for both GET and POST HTTP Method Requests.
A	The service SHALL support both GET and POST HTTP Methods for queries.

ANNEX A
(NORMATIVE)
ABSTRACT TEST SUITE

CONFORMANCE CLASS A.1	
IDENTIFIER	`http://www.dgiwg.org/std/edr/1.0/conf/nwp_radar`
REQUIREMENTS CLASS	Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_radar`
CONFORMANCE TESTS	Abstract test A.1: `/conf/nwp_radar/spatial-extents-crs` Abstract test A.2: `/conf/nwp_radar/profile-bbox-coordinates` Abstract test A.3: `/conf/nwp_radar/profile-trs-values` Abstract test A.4: `/conf/nwp_radar/profile-vrs-values` Abstract test A.5: `/conf/nwp_radar/profile-keywords` Abstract test A.6: `/conf/nwp_radar/profile-links` Abstract test A.7: `/conf/nwp_radar/profile-query-types` Abstract test A.8: `/conf/nwp_radar/items-endpoint` Abstract test A.9: `/conf/nwp_radar/cube-endpoint` Abstract test A.10: `/conf/nwp_radar/radius-endpoint` Abstract test A.11: `/conf/nwp_radar/profile-nwp-parameters` Abstract test A.12: `/conf/nwp_radar/http-methods`

ABSTRACT TEST A.1	
IDENTIFIER	`/conf/nwp_radar/spatial-extents-crs`
REQUIREMENT	Requirement 1: `/req/nwp_radar/spatial-extents-crs`
TEST PURPOSE	Validate that spatial extents crs is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation Verify that the bbox geometry is defined in WGS84. Verify that the bounding box coordinates are presented as geographical latitude, then longitude.

ABSTRACT TEST A.1	
	Verify that the bounding box coordinates are returned in decimal degrees.

ABSTRACT TEST A.2	
IDENTIFIER	`/conf/nwp_radar/profile-bbox-coordinates`
REQUIREMENT	Requirement 2: `/req/nwp_radar/profile-bbox-coordinates`
TEST PURPOSE	Validate that profile bbox coordinates is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation Verify that the box boundary is for the extent [-180, -90, 180, 90]. Verify that the bounding box coordinate reference system is EPSG:4326 (WGS84).

ABSTRACT TEST A.3	
IDENTIFIER	`/conf/nwp_radar/profile-trs-values`
REQUIREMENT	Requirement 3: `/req/nwp_radar/profile-trs-values`
TEST PURPOSE	Validate that profile trs values is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation Verify that the trs is in Gregorian calendar Zulu datetime (UTC) in 24 hour clock with a precision to the second. Verify that the trs is in the time format specified in RFC 3339 e.g. 2026-04-12T21:20:50.52Z.

ABSTRACT TEST A.4	
IDENTIFIER	`/conf/nwp_radar/profile-vrs-values`
REQUIREMENT	Requirement 4: `/req/nwp_radar/profile-vrs-values`

ABSTRACT TEST A.4	
TEST PURPOSE	Validate that profile vrs values is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation Verify that the vrs is defined in height metres. Verify that the vrs is defined as WKT2 EPSG:3855. Verify that the vrs has the vertical datum EGM2008 geoid as the zero-height reference. Verify that extent.vertical.positive is present and is either “up” or “down”.

ABSTRACT TEST A.5	
IDENTIFIER	`/conf/nwp_radar/profile-keywords`
REQUIREMENT	Requirement 5: `/req/nwp_radar/profile-keywords`
TEST PURPOSE	Validate that profile keywords is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation Verify that the collection keywords list includes the following keywords — “cube”, “radius”, “items”, “precipitation_amount”, “radar”.

ABSTRACT TEST A.6	
IDENTIFIER	`/conf/nwp_radar/profile-links`
REQUIREMENT	Requirement 6: `/req/nwp_radar/profile-links`
TEST PURPOSE	Validate that profile links is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation

ABSTRACT TEST A.6	
	Verify that the links object consists of href, rel, title and type elements and that they are filled in with appropriate values. Send GET request to the root URL / Verify that the links object consists of href, rel, title and type elements and that they are filled in with appropriate values.

ABSTRACT TEST A.7	
IDENTIFIER	`/conf/nwp_radar/profile-query-types`
REQUIREMENT	Requirement 7: `/req/nwp_radar/profile-query-types`
TEST PURPOSE	Validate that profile query types is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation Verify that the following data queries are listed — “cube”, “items”, “radius”.

ABSTRACT TEST A.8	
IDENTIFIER	`/conf/nwp_radar/items-endpoint`
REQUIREMENT	Requirement 8: `/req/nwp_radar/items-endpoint`
TEST PURPOSE	Validate that items endpoint is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation/items. Verify response Content-Type is application/json. Verify each feature contains precipitation_amount property. Verify each feature contains datetime property. Verify that the datetime is in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z Verify that the default file format is GeoTIFF.

ABSTRACT TEST A.8	
	Verify that the endpoint supports CoverageJSON FeatureCollection and GRIB2, GeoTIFF, HDF5, NetCDF4, PNG file formats. Send GET request to /collections/nwp_radar_precipitation/items for a CoverageJSON file type. Verify that the CoverageJSON file format returned complies with the OGC CoverageJSON standard — https://docs.ogc.org/cs/21-069r2/21-069r2.html#_df6ce66a-7369-49cb-b59f-47a8c5d424ca Send GET request to /collections/nwp_radar_precipitation/items for a GeoTIFF file type. Verify that the GeoTIFF file format returned complies with the DGIWG 108 GeoTIFF for GeoRef Imagery Standard. Send GET request to /collections/nwp_radar_precipitation/items for a GRIB file type. Verify that the GRIB2 file format returned complies with the WMO Manual On Codes — https://library.wmo.int/records/item/35625-manual-on-codes-volume-i-2-international-codes. Send GET request to /collections/nwp_radar_precipitation/items for a HDF5 file type. Verify that the HDF5 file format returned complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html. Send GET request to /collections/nwp_radar_precipitation/items for a NetCDF4 file type. Verify that the NetCDF4 file format returned complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.

ABSTRACT TEST A.9	
IDENTIFIER	`/conf/nwp_radar/cube-endpoint`
REQUIREMENT	Requirement 9: `/req/nwp_radar/cube-endpoint`
TEST PURPOSE	Validate that cube endpoint is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation/cube with a valid coordinate in CRS EPSG:4326. Verify response Content-Type is application/json.

ABSTRACT TEST A.9

Verify each feature contains precipitation_amount property.

Verify each feature contains datetime property.

Verify each feature has height in metres based upon the EGM2008 gravity model.

Verify that the datetime is in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z

Verify that the default file format is GeoTIFF.

Verify that the endpoint supports CoverageJSON FeatureCollection and GRIB2, GeoTIFF, HDF5, NetCDF4, PNG file formats.

Send GET request to /collections/nwp_radar_precipitation/cube for a CoverageJSON file type.

Verify that the CoverageJSON file format returned complies with the OGC CoverageJSON standard — https://docs.ogc.org/cs/21-069r2/21-069r2.html#_df6ce66a-7369-49cb-b59f-47a8c5d424ca

Send GET request to /collections/nwp_radar_precipitation/cube for a GeoTIFF file type.

Verify that the GeoTIFF file format returned complies with the DGIWG 108 GeoTIFF for GeoRef Imagery Standard.

Send GET request to /collections/nwp_radar_precipitation/cube for a GRIB file type.

Verify that the GRIB2 file format returned complies with the WMO Manual On Codes — https://library.wmo.int/records/item/35625-manual-on-codes-volume-i-2-international-codes.

Send GET request to /collections/nwp_radar_precipitation/cube for a HDF5 file type.

Verify that the HDF5 file format returned complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.

Send GET request to /collections/nwp_radar_precipitation/cube for a NetCDF4 file type.

Verify that the NetCDF4 file format returned complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.

ABSTRACT TEST A.10

IDENTIFIER	`/conf/nwp_radar/radius-endpoint`
REQUIREMENT	Requirement 10: `/req/nwp_radar/radius-endpoint`

ABSTRACT TEST A.10	
TEST PURPOSE	Validate that radius endpoint is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation/radius with a valid coordinate in EPSG:4326 and a radius of 1km. Verify response Content-Type is application/json. Verify each feature contains precipitation_amount property. Verify each feature contains datetime property. Verify each feature has height in metres based upon the EGM2008 gravity model. Verify that the datetime is in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z Verify that the default file format is GeoTIFF. Verify that the endpoint supports CoverageJSON FeatureCollection and GRIB2, GeoTIFF, HDF5, NetCDF4, PNG file formats. Send GET request to /collections/nwp_radar_precipitation/radius for a CoverageJSON file type. Verify that the CoverageJSON file format returned complies with the OGC CoverageJSON standard — https://docs.ogc.org/cs/21-069r2/21-069r2.html#_df6ce66a-7369-49cb-b59f-47a8c5d424ca Send GET request to /collections/nwp_radar_precipitation/radius for a GeoTIFF file type. Verify that the GeoTIFF file format returned complies with the DGIWG 108 GeoTIFF for GeoRef Imagery Standard. Send GET request to /collections/nwp_radar_precipitation/radius for a GRIB file type. Verify that the GRIB2 file format returned complies with the WMO Manual On Codes — https://library.wmo.int/records/item/35625-manual-on-codes-volume-i-2-international-codes. Send GET request to /collections/nwp_radar_precipitation/radius for a HDF5 file type. Verify that the HDF5 file format returned complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html. Send GET request to /collections/nwp_radar_precipitation/radius for a NetCDF4 file type.

ABSTRACT TEST A.10	
	Verify that the NetCDF4 file format returned complies with the CF-NetCDF v1.13 conventions — https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html.

ABSTRACT TEST A.11	
IDENTIFIER	`/conf/nwp_radar/profile-nwp-parameters`
REQUIREMENT	Requirement 11: `/req/nwp_radar/profile-nwp-parameters`
TEST PURPOSE	Validate that profile nwp parameters is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/nwp_radar_precipitation/ Verify that datetime is formatted in Gregorian calendar Zulu datetime (UTC) in 24 hour clock with a precision to the second. Verify that the datetime format is aligned to RFC 3339. Verify that precipitation_amount uses kg m-2 units.

ABSTRACT TEST A.12	
IDENTIFIER	`/conf/nwp_radar/http-methods`
REQUIREMENT	Requirement 12: `/req/nwp_radar/http-methods`
TEST PURPOSE	Validate that http methods is correctly implemented.
TEST METHOD	
STEP	Send GET request to the root URL / Verify that the service is set up to support both POST and GET HTTP Method requests. Verify that POST and GET Methods are supported for the query types of “cube”, “radius”, and “items”.